
TEAM HARRIS AND KIELO MODEL REPORT *

Matthew Kielo
mkielo3@gatech.edu

Harris Alterman
harrisjayalterman@gmail.com

ABSTRACT

This paper describes our 2nd place solution for the PREPARE Acoustic Challenge, which detects Mild Cognitive Impairment and Alzheimer’s Disease and Related Dementias (ADRD) using speech biomarkers. Our model combines three sub-models: eGeMAPS-v2 acoustic features, and two Whisper fine-tunes (one using triplet loss to control for demographics). The ensemble achieves an AUROC of 0.85 and demonstrates strong generalization across populations, with post-hoc analysis revealing interpretable error patterns and areas for improvement.

1 Introduction

The holy grail of acoustic ADRD detection is an algorithm that can process a short clip of anyone’s speech, devoid of all other context, and accurately assign a diagnosis. That is not this solution. Instead, we attempt to answer the following, much easier, research questions:

1. Can acoustic biomarkers be used to outperform a non-acoustic baseline for the prediction of ADRD?
2. How do these models compare across: 1) prediction accuracy, 2) bias mitigation, 3) generalization to new patients within studied populations, and 4) generalization to underrepresented populations?

We hope these answers will contribute a small increment of progress towards the ambitious ultimate goal.

2 Methodology

Speech contains two broad types of information: *lexical* (what was said) and *non-lexical* (how it was said.) Our final model combines three submodels that process different combinations of these elements. The submodels reuse 3 common components: (1) *XGBoost Classifier* (XGB) [1], a gradient boosted decision tree ensemble. We use XGB to classify tabular data. (2) *Whisper*, [2] a transcription model that encodes multi-lingual audio to vectors and decodes it to text. We use Whisper-Large-V3 for transcriptions, and Whisper-Small as a base model. (3) *5-Fold Cross Validation* segments the dataset into 5 folds. Each fold serves once as validation while others train. We use 5-fold CV to mitigate overfitting.

2.1 Preprocessing

2.1.1 Semantic Clustering

Goal. Cluster audio by lexical content to control for different sampling protocols. Cluster provides a feature for the non-acoustic baseline and provides a factor to evaluate generalization across cohorts.

Implementation. Using only the training set: 1) Transcribe audio with Whisper-Large-V3, 2) Embed using all-MiniLM-L6-v2 sentence transformer [3], 3) Project to two dimensions with UMAP [4], 4) Cluster with HDBScan [5], 5) Train XGB to map UMAP embeddings to HDBScan cluster. Cluster the test set using the resulting trained models.

Results. Found 5 clusters which we treat as metadata alongside age and gender. Post-hoc analysis reveals cluster to be a subset of corpus with the following associations: 0: [Ye], 1: [Baycrest, Delaware, DePaul, Hopkins, Lu, Pitt, WLS], 2: [VAS], 3: [Ivanova], 4: [Ivanova].

*Training code will be released to https://github.com/mkielo3/PREPARE_acoustic, weights available to organizers upon request. All dependencies are confirmed to be fully open-sourced.

Limitations. (1) Cluster 1 contains 75% of samples, suggesting potential for finer clustering. (2) Model requires in-distribution data for reliable clustering. Out-of-distribution samples can be marked as unknown.

2.1.2 Non-Acoustic Baseline

Goal. Establish a non-acoustic baseline (NAB) for comparison against acoustic models.

Implementation. XGB is trained on age, gender, and semantic cluster to predict diagnosis.

Limitations. Limited by available metadata, but could be improved by population base-rates in practice.

2.2 Model Descriptions

2.2.1 Submodel 1: XGB trained with eGeMAPS-v2 Features

Goal. Predict impairment using eGeMAPS-v2 acoustic descriptors.

Implementation. 1) Aggregate the 88 pre-generated, non-lexical acoustic features by patient with mean, standard deviation, minimum, maximum, median, inter-quartile mean, and inter-quartile range, resulting in 616 features. 2) Add metadata (age, gender, semantic cluster) 3) Fit XGB using 5-fold cross validation and multi-class log loss.

Limitations. 1) Diagnosis may be confounded by age, gender, and race. Including metadata makes bias explicit. 2) Aggregating to summary statistics may lose important structure.

Additional Details. Feature importance is measured by *gain*. Gain counts the features used to split the data, weighted by improvement in accuracy, normalized to 1. The 10 highest gain features contain: 5 related to loudness, 3 binary indicators of cluster assignment, and 2 related to voice quality. The 7 numeric features use robust aggregations (IQR, median, or IQM). The feature importance distribution has a low Gini coefficient of 0.2, indicating predictions rely on a wide variety of features.

2.2.2 Submodel 2: Fine Tuning Whisper with a Classification Head

Goal. Predict impairment using a foundation model.

Implementation. Add 3 blocks of (linear layer, ReLU, and 20-30% dropout) to the output of the Whisper-Small encoder. Freeze Whisper layers and train with 5-fold cross validation while augmenting samples with Gaussian noise (0.0005 to 0.002). Optimize multi-class log loss for 25-50 epochs, with early stopping using a 5 epoch average on the validation set. Training typically takes 30-40 epochs.

Limitations. 1) Outputs are difficult to interpret. 2) Biased towards lexical content.

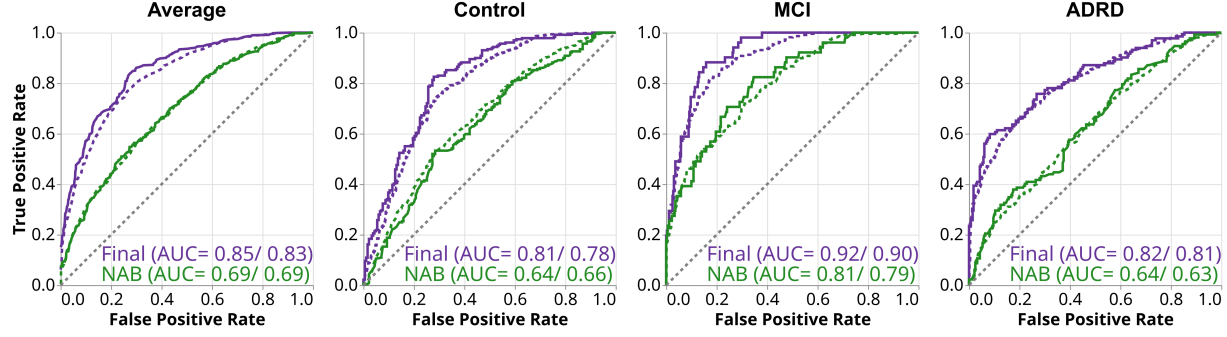
Additional Details. 1) Whisper-Small performs best compared to other sizes. 2) Unfreezing any Whisper layers causes rapid overfitting. 3) 3-layer MLP outperforms 2-layer and CNN variants. 4) We train two models, using 10 second and 30 second audio lengths respectively. The final prediction averages both models. Right-zero padding is applied as necessary. The final output averages the prediction from both models. 5) Additional forms of augmentation did not improve performance.

2.2.3 Submodel 3: Whisper Classifier with Triplet Loss

Goal. Develop Whisper-based classifier, made invariant to age, gender, and semantic content.

Implementation. We train an architecture similar to submodel 2 but with triplet loss [6] (a metric that directly compares similar samples that differ in class, requiring a hidden layer to embed the same class samples close-by while pushing opposite classes apart, despite forced similarities). Specifically we: 1) Combine MCI and ADRD diagnoses resulting in a binary classification problem. 2) Select samples sharing demographic and semantic features but differing in diagnosis, requiring: (1) same gender (2) ages within 10 years (3) same semantic cluster (4) similar lexical content. 3) Build triplet bank containing anchor, positive example, negative example based on these matches. 4) Optimize triplet loss and classification loss.

Limitations. 1) Poor triplet bank construction may introduce unknown biases. 2) Binary classification loses granularity. 3) Exogenous factor control may hinder performance.

a. Final Model Receiver Operating Characteristic (ROC) by Diagnosis**b. Confusion Matrix (Test)**

True Class	Predicted Class		
	Control	MCI	ADRD
Control	205 (73.5%)	10 (23.8%)	14 (15.4%)
MCI	19 (6.8%)	26 (61.9%)	6 (6.6%)
ADRD	55 (19.7%)	6 (14.3%)	71 (78.0%)

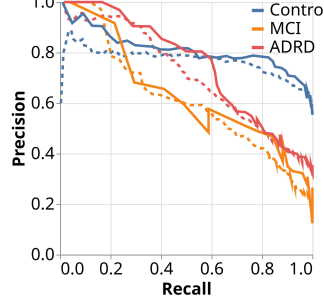
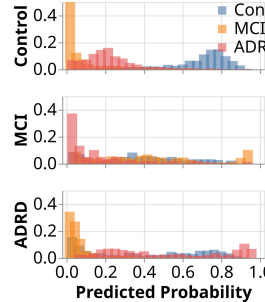
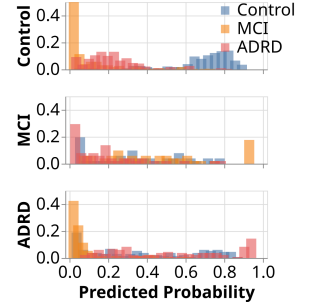
c. Precision-Recall Curve**d. Predictions by Class (Train)****e. Predictions by Class (Test)**

Figure 1: **a.** ROC by disease. Solid/dashed lines indicate test/train set respectively. **b.** Confusion matrix of accuracy by class. **c.** Precision-recall curves. Solid/dashed lines indicate test/train set respectively. **d, e.** Distribution of predicted class by true class for train/test sets.

Additional Details. The embedding space shows separation between diagnostic classes, suggesting successful identification of relevant features while controlling for confounding variables. Note: the final model does not require triplets for inference.

2.2.4 Combining the Final Model

Goal. Create a final composite prediction using the output of each submodel.

Implementation. The validation fold predictions are aggregated across all submodels. The final feature set contains: three submodel 1 class probabilities, three submodel 2 class probabilities, one sub submodel 3 class probability along with age, and semantic cluster. The final model is trained with XGB and 5-fold cross validation. Test time augmentation is applied to submodels 2 and 3, with the final prediction containing the average of the original test sample with 5 distorted copies (using Gaussian noise 0.0005-0.002). This aggregation is empirically confirmed to modestly improve performance.

Notable Features. Low-Gini coefficient (0.075) across features indicates evenly distributed predictive power. Feature gain shows highest importance for sub-model 2 (HC: 0.16, MCI: 0.15, ADRD: 0.17), a mix of sub-model 1 (HC: 0.11, MCI: 0.07, ADRD: 0.07) and sub-model 3 (HC: 0.1), then metadata (cluster: 0.08, age: 0.08). Gender did not improve performance.

3 Model Performance

Overall Performance. The model achieves a private leaderboard score of 0.634, ranking #2, inline with its local CV of 0.664 ± 0.033 . The ROC curve of predictions are presented in Fig 1. Clinicians commonly rely on ROC to evaluate the discriminatory power of a model for different clinical settings. **Key results:** 1) AUROC of 0.85 on the test set, significantly outperforming the NAB across all classes. 2) Train/test accuracies of 69%/73%. 3) Consistent predictions across train and test sets based on bootstrapped Kolmogorov-Smirnov tests ($KS = 0.05$, $p = 0.37$) and Brier Score tests (mean difference = 0.03, $p = 0.50$), with 10,000 trials. This indicates no evidence of overfitting and crucially allows us to aggregate train and test sets for post-hoc analysis, strengthening conclusions given limited data.

Bias Analysis. Table 1 examines model bias by sub-population, with particular attention to young pre-retirement subgroups who have potential for early diagnosis. While the model significantly outperforms the NAB baseline across all groups, there are notable variations in performance. **Key results:** 1) the final model achieves the highest AUROC for 14/15 demographic groups, 2) most sub-models improve over baseline, 3) comparatively weak performance for Spanish speakers, 4) strong results for high-risk combinations, particularly young females, though young Spanish speakers continue to show lower performance, and 5) strong performance for African Americans despite poor NAB scores, though limited by very small sample size.

Table 1: **Bias Analysis by Demographic.** AUROC scores across population subgroups and model variants. Combines train and test sets while requiring groups to have 2+ diagnostic classes (excluding Mandarin speakers who only contain MCI diagnoses). Intersectional analysis focuses on young, high-risk subgroups to emphasize early detection potential.

		Final	Sub-1	Sub-2	Sub-3	NAB	Count
Age	Young (69 and below)	0.87*	0.77*	0.87*	0.83*	0.65	522
	70 to 79	0.84*	0.80*	0.83*	0.76*	0.61	606
	80 and Older	0.78*	0.76*	0.77*	0.74	0.73	930
Education	Incomplete Secondary	0.75*	0.73*	0.74*	0.72*	0.66	262
	Completed Secondary	0.76*	0.74	0.74*	0.72	0.71	546
	Higher Education	0.83*	0.79*	0.81*	0.76*	0.67	525
Gender	Female	0.84*	0.79*	0.84*	0.80*	0.71	1,219
	Male	0.81*	0.76*	0.80*	0.75*	0.67	839
Language	English	0.84*	0.78*	0.83*	0.78*	0.67	1,655
	Spanish	0.69*	0.68*	0.67*	0.65*	0.60	360
Race	African American	0.92*	0.90*	0.74*	0.80*	0.39	9
	White	0.76*	0.73*	0.85*	0.81*	0.54	99
Intersectional	Young, Female	0.89*	0.78*	0.88*	0.84*	0.64	311
	Young, Incomplete Secondary Ed.	0.85*	0.85*	0.83	0.85*	0.68	55
	Young, Spanish Language	0.67*	0.64*	0.61	0.58	0.50	75

(*) indicates statistical significance vs. NAB baseline ($p < 0.05$, 1,000 bootstrap samples).

Error Analysis by Factor. Model errors are interacted with a set of interpretable factors, summarized in Table 2. The factors are organized into 3 groups: 1) examines basic acoustic features, 2) basic linguistic features of the transcript (with readability corresponding to predicted grade level [7]), and 3) Whisper based emotion classification [8]. High emotional entropy indicates no dominant emotion. Error analysis is performed to identify errors recognizing impairment (Control or MCI/ADRD, across the full population) or severity (MCI or ADRD, across impaired population). Error is calculated as $1 - \hat{y}$ for the correct class and is always positive. A positive correlation indicates underdiagnosis (higher error) in the presence of the factor, while a negative correlation indicates underdiagnosis in its absence. Due to the binary classification setting, underdiagnosis in one category implies overdiagnosis in the other. **Key results:** 1) Control is underdiagnosed in loud recordings, 2) MCI is underdiagnosed in transcripts with simple language, 3) ADRD is underdiagnosed in samples without a dominant emotion (high emotional entropy), 4) Control is underdiagnosed in recordings with high anger.

Error analysis by corpus is detailed in 3. The final model consistently outperforms the NAB baseline. However, the notably strong performance on the Ye dataset should be interpreted with caution. This dataset consists exclusively of Mandarin speakers with MCI, the extremely low log-loss suggests the model is simplistically associating Mandarin speech with MCI.

Generalization: Our model demonstrates strong in-distribution generalization through both design choices and empirical results. Structurally: (1) 5-fold cross validation with early stopping limits overfitting. (2) Ensemble architecture prevents over-reliance on single factors. (3) High dropout rates (20-30%) are used in deep learning components. Empirically we see: (4) Consistent train/test performance and error distributions. (5) Strong performance across semantic clusters (Table 4). (6) Robust performance across demographics (Table 1.).

Table 2: **Error Analysis by Factor.** Correlation Coefficients between Error and Factor.

Feature		Full Population Control MCI or ADRD		Impaired Only MCI ADRD	
Acoustic	Loudness	0.21*	-0.05	-0.29*	0.07
	Brightness	-0.03	-0.09*	0.13	0.15*
	Clarity	0.12*	-0.10*	0.08	0.14*
	Duration	-0.02	0.08	0.39*	0.06
Linguistic	Word Count	-0.19*	0.31*	0.03	-0.13*
	Readability	0.09*	0.04	-0.28*	0.21*
Emotion	Anger	0.21*	-0.11*	-0.08	0.16*
	Disgust	-0.02	-0.07	-0.11	0.1
	Fear	0.12*	-0.14*	-0.14	0.03
	Happy	-0.11*	0.06	0.19*	-0.21*
	Neutral	-0.09*	0.17*	0.11	0.13*
	Sad	-0.07	-0.12*	-0.02	-0.07
	Surprise	0.00	0.15*	-0.03	0.01
	<i>Entropy</i>	0.10*	0.02	-0.04	0.23*
Count		1,140	918	268	650

(*) indicates statistical significance. Rank correlations are used.

Table 3: **Error Analysis of Final Model Log Loss by Corpus.**

	Log Loss		Count	
	Train	Test	Train	Test
Baycrest	0.59*	0.63*	16	4
Delaware	0.78*	0.83*	68	13
DePaul	0.12*		1	0
Hopkins	0.26*	0.22*	24	4
Ivanova	0.93	0.90	281	79
Lu	0.70*	0.88	37	14
Pitt	0.69*	0.47*	439	110
Vas	0.77*	0.95	86	14
WLS	0.58*	0.59*	660	165
Ye	0.08*	0.07*	34	9

(*) indicates statistical significance over NAB, calculated with $p < 0.05$ over 1,000 bootstrapped samples.

To assess out of distribution generalization, we re-train the model using careful curated variants of the dataset, shown in 2. In this study, the training set is constructed to include 0, 10, or 50 samples from the simulated out of distribution dataset. Model results are calculated and compared to a model relying on metadata predictions. This analysis tests Vas (remote collection) and Ivanova (Spanish language) datasets. These results show that an acoustic model can improve over the baseline out of distribution, even with limited samples.

Explainability. Surrogate models are trained to better understand model predictions on English-speaking women under 70 ($n=243$; 147 control, 23 MCI, 73 ADRD). The group is chosen to limit confounding variables while studying early detection in a high-risk population. The interpretable feature set combines eGeMAPS-v2 acoustic aggregations and error analysis factors, resulting 631 variables. The features are clipped and z-scored, then transformed with principal component analysis. The first 25 PCs explain 80% of variance, emphasizing the data’s high dimensionality. Next, we identify the best fitting two-PC linear regression models to predict each submodel output, summarized in Table 3. The selected PCs are manually annotated guided by their correlations to the underlying features. The surrogate model has much weaker downstream predictive ability than the final model, but does outperform the NAB, shown in Table 3. **Key results:** 1) Vocal resonance (PC1) detects different stages of cognitive decline, increasing with MCI but decreasing for ADRD. 2) Voice clarity (PC3) and timbre variations (PC5) emerge as important markers, suggesting that subtle changes in speech patterns could serve as early indicators of cognitive status across diagnosis. 3) High errors for submodels 2 and 3 suggest Whisper captures non-linear information inaccessible

Table 4: **Train/Test Performance** by Model and Training Cluster.

		Train					Test				
		All	C=1	C=2	C=3	C=4	All	C=1	C=2	C=3	C=4
AUROC	Final	0.83*	0.83*	0.77*	0.72*	0.59*	0.85*	0.86*	0.80*	0.62*	0.81*
	Sub-1	0.78*	0.76*	0.72*	0.69*	0.62*	0.78*	0.78*	0.55	0.56	0.69
	Sub-2	0.82*	0.81*	0.78*	0.69*	0.59	0.85*	0.85*	0.75*	0.65*	0.80*
	Sub-3	0.77*	0.74*	0.66*	0.60	0.55	0.81*	0.78*	0.46	0.58	0.70
	NAB	0.69	0.64	0.49	0.57	0.53	0.72	0.68	0.47	0.51	0.57
Log Loss	Final	0.68*	0.63*	0.75*	0.95*	0.90	0.64*	0.57*	0.94	1.09	0.72*
	Sub-1	0.76*	0.73*	0.88*	0.97*	0.90	0.76*	0.71*	1.09	1.20	0.79
	Sub-2	0.69*	0.64*	0.72*	0.97*	0.91	0.64*	0.59*	0.92*	1.06	0.72*
	Sub-3	0.76*	0.71*	0.98*	1.08	0.90	0.72*	0.66*	1.12	1.11	0.76*
	NAB	0.85	0.83	1.10	1.05	0.89	0.84	0.81	1.15	1.12	0.86
Count		1,646	1,243	88	130	150	412	310	14	38	41

NAB is Non-Acoustic Baseline. C=1 to C=4 represent clusters 1 to 4. Cluster 0 removed for insufficient class representation. (*) indicates statistical significance over NAB, calculated with $p < 0.05$ over 1,000 bootstrapped samples.

Out of Distribution Generalization

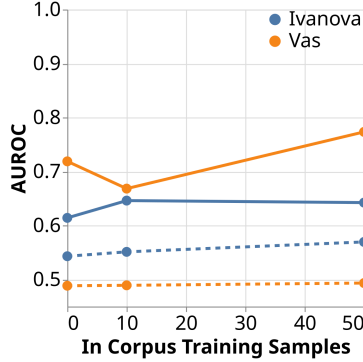


Figure 2: Shows model performance in a simulated out of distribution setting. Full model shown in solid line, with corresponding NAB (excluding cluster) in dashed. The full training procedure is repeated, using the full dataset, but only a limited number of samples from the out-of-distribution corpus (indicated on the x-axis). Tests show the both simulations beat a corresponding age/gender only baseline.

to the PCs. 4) All selected PCs correspond to eGeMAPS-v2 acoustic factors, indicating their significance relative to the error factors.

Table 5: **Explainability Analysis**. Optimal 2-PC surrogate linear regression models by sub-predictor.

Target Factor	Model	SSE	SE
Sub-1: HC	$-0.012 \times PC7 + 0.010 \times PC3 + 0.524$	3.78	0.13
Sub-1: MCI	$-0.013 \times PC3 + 0.006 \times PC1 + 0.143$	2.43	0.10
Sub-1: AD RD	$-0.014 \times PC5 + -0.005 \times PC1 + 0.333$	4.13	0.13
Sub-2: HC	$0.014 \times PC8 + -0.014 \times PC7 + 0.527$	13.38	0.24
Sub-2: MCI	$-0.012 \times PC3 + 0.007 \times PC1 + 0.114$	4.99	0.14
Sub-2: AD RD	$-0.018 \times PC5 + -0.005 \times PC1 + 0.359$	14.79	0.25
Sub-3: HC	$-0.014 \times PC7 + 0.012 \times PC5 + 0.543$	10.09	0.20

Principal components are manually identified to correspond to: PC1 (resonance strength/timing), PC3 (voice clarity), PC5 (timbre variations), PC7 (pitch expressiveness), PC8 (voice power control)

Table 6: **Performance Metrics of Surrogate Model.**

Model	Log Loss	AUROC
NAB	0.89	0.51
Final	0.54	0.88
Surrogate	0.82	0.68

4 Conclusion

"Constructing a predictive model is easy, but authenticity and practical use are difficult" - Edward Tufte.

Our model demonstrates robust performance, consistently outperforming non-acoustic baselines across diverse populations, however there is significant room for future work. In particular: (1) Clinical Integration: Collaboration with healthcare providers is essential to understand how this technology can best serve clinical needs, whether as a screening tool, a supplement to existing diagnostics, or a measure of disease progression. Neither author has experience in the healthcare field and would both benefit tremendously from domain expertise. (2) Enhanced Interpretability: While our surrogate model approach demonstrates the potential for explaining black-box predictions, there is significant room to incorporate additional linguistic and prosodic features into these explanations and to develop population-specific interpretable models, as well as improve performance of the surrogate models. (3) Technical Improvements: Current limitations in the speech processing pipeline, particularly around emotional content preservation, suggest opportunities for incorporating new language models designed for emotion-aware speech processing [9]. (4) Data Collection Strategy: Our out-of-distribution analysis provides a framework for strategic data collection, particularly for underrepresented populations like Mandarin speakers. Understanding minimum data requirements for reliable model adaptation could guide future research efforts. Even additional control data would help further analysis. (5) Bias Mitigation: Though our model shows promising cross-population performance, continued work on fairness and bias mitigation remains crucial for clinical applications

References

- [1] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. 2016.
- [2] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision, 2022.
- [3] Sentence transformer: all-minilm-l6-v2. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>. Accessed: 2024-12-15.
- [4] Leland McInnes, John Healy, Nathaniel Saul, and Lukas Grossberger. Umap: Uniform manifold approximation and projection. *The Journal of Open Source Software*, 3(29):861, 2018.
- [5] Leland McInnes, John Healy, and Steve Astels. hdbscan: Hierarchical density based clustering. *The Journal of Open Source Software*, 2(11):205, 2017.
- [6] Florian Schroff, Dmitry Kalenichenko, and James Philbin. Facenet: A unified embedding for face recognition and clustering. 2015.
- [7] Deberta v3 base for multilingual readability assessment. [agentlans/mdeberta-v3-base-readability](https://huggingface.co/agentlans/mdeberta-v3-base-readability). Accessed: 2025-01-05.
- [8] Speech emotion recognition with whisper. <https://huggingface.co/firdhokk/speech-emotion-recognition-with-openai-whisper-large-v3>. Accessed: 2025-01-05.
- [9] Yu-An Chung Mariano Cora Meglioli David Dale Ning Dong Paul-Ambroise Duquenne Hady Elsahar Hongyu Gong Kevin Heffernan John Hoffman Christopher Klaiber Pengwei Li Daniel Licht Jean Maillard Alice Rakotoarison Kaushik Ram Sadagopan Guillaume Wenzek Ethan Ye Bapi Akula Peng-Jen Chen Naji El Hachem Brian Ellis Gabriel Mejia Gonzalez Justin Haaheim Prangthip Hansanti Russ Howes Bernie Huang Min-Jae Hwang Hirofumi Inaguma Somya Jain Elahe Kalbassi Amanda Kallet Ilia Kulikov Janice Lam Daniel Li Xutai Ma Ruslan Mavlyutov Benjamin Peloquin Mohamed Ramadan Abinеш Ramakrishnan Anna Sun Kevin Tran Tuan Tran Igor Tufanov Vish Vogeti Carleigh Wood Yilin Yang Bokai Yu Pierre Andrews Can Balioglu Marta R. Costa-jussà³ Onur Celebi Maha Elbayad Cynthia Gao Francisco Guzmán Justine Kao Ann Lee Alexandre Mourachko Juan Pino Sravya Popuri Christophe Ropers Safiyyah Saleem Holger Schwenk Paden Tomasello Changan Wang Jeff Wang Skyler Wang Seamless Communication, Loïc Barrault. "seamlessm4t—massively multilingual & multimodal machine translation". *ArXiv*, 2023.